

Ben Nguyen

(239) 888-0663 | benn5334@gmail.com | github.com/HuuBen5334 | linkedin.com/in/nguyenhuuben | Portfolio

EDUCATION

University of Florida

B.S. Computer Engineering GPA: 3.65

Gainesville, FL

Expected Dec 2027

Coursework: Data Structures & Algorithms, Microprocessor Applications, Digital Logic, Circuits I, AI for Virtual Environments

TECHNICAL SKILLS

Languages: C, C++, C#, Java, Python, JavaScript, AVR Assembly, VHDL, SQL

Embedded & Hardware: ESP32, Arduino, Altium Designer, I2S, Quartus Prime, SolidWorks

Software & Frameworks: Spring Boot, React, gRPC, REST APIs, WebSockets, PostgreSQL, Godot, Git

Certificates: Siemens Expedite Microcredential, AutoCAD Inventor

PROJECTS

Portable Bluetooth Speaker – *ESP32, C++, Altium Designer, I2S, SolidWorks, PCB Design*

- Designed a custom 2-layer PCB in Altium Designer that integrates dual MAX98357A class-D amplifiers. Connected to an ESP32 microcontroller that is powered by a portable battery bank.
- Developed C++ firmware (Arduino framework) streaming Bluetooth audio over the I2S digital protocol to dual 5W speakers, with hardware interrupts and debouncing for tactile play/pause push-buttons.
- Programmed an LCD to render real-time system status and playback metadata; modeled and 3D-printed an enclosure to house the PCB, speakers, display, and controls using SolidWorks.

Full-Stack E-Commerce Marketplace – *Java, Spring Boot, C++, gRPC, React, PostgreSQL*

- Designed and developed an e-commerce marketplace using React, Spring Boot, and PostgreSQL, featuring a custom C++ remote procedure call (gRPC) microservice for real-time dynamic pricing calculations.
- Eliminated a 96% order failure rate during high-traffic load testing by implementing optimistic locking.
- Delivered real-time order notifications via STOMP-over-WebSocket, validated the system with Mockito unit tests.

Robocode Tournament AI – *Java, Robocode API, Finite State Machines, Minimum-Risk Movement*

- Placed 1st among 26 competitors with an autonomous Java tank AI, outscoring the benchmark bot by 91%; a finite-state machine (scan/attack/evade) governed target selection, firepower, and movement policy.
- Built a minimum-risk movement engine scoring 300 sampled destinations per re-position against an inverse-square risk field weighted by relative enemy energy and firing angle, minimizing exposure to incoming fire.
- Dodged incoming shots by inferring enemy fire from per-tick energy deltas and inverting strafe direction; led targets using a linear-intercept solution derived from bullet velocity and enemy heading.

FishSchooling Simulation – *C#, Godot Engine, Git*

- Implemented flocking (alignment, cohesion, separation) on an interface-based steering system to model autonomous 2D agents; collaborated with 3 developers via Git branching and pull requests.
- Built a dynamic UI exposing fish-type-specific parameter sliders, enabling real-time behavior tuning across 5 creature types in a predator-prey ecosystem.

EXPERIENCE

UF Association of Computer Engineers

Workshop Lead

Gainesville, FL

March 2026 – Present

- Led brainstorming of events that provided opportunities to learn about computer engineering and its applications.
- Improved club engagement by hosting research lab tours and technical workshops; coordinated with presenters and staff to ensure seamless event execution.

Kyoto University — Cross-Cultural Engineering Design Program

Exchange Student & Networking Event Coordinator

Kyoto, Japan

September 2025 – December 2025

- Selected for a semester exchange; designed on mixed Japanese-international teams to communicate technical concepts to users with no shared language or cultural reference, producing FishSchooling.
- Implemented pathfinding and decision-making models to drive autonomous agent behavior in *AI for Virtual Env.*
- Organized lunch networking events averaging 60 attendees with activities to facilitate cross-cultural interaction.
- Increased Japanese student attendance by 40% over prior semesters through outreach and campus-wide promotion.